

Analysis of Gamma Emissions from Eight Beta Decay Contributors to the Reactor

Antineutrino Spectrum Anomaly

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Abstract

The Reactor Anti-Neutrino Anomaly (RAA) is a bump in the reactor antineutrino spectrum around 5 MeV which is not predicted by standard nuclear data, and which disrupts the interpretation of neutrino oscillation experiments. These experiments are especially important in probes of lepton conservation law violation and fundamental particle physics. Neutrinos are elementary particle, produced through radioactive interactions, with no electrical charge, travels nearly the speed of light, passes through ordinary matter with virtually no interaction and are produced in huge numbers by nuclear reactors. ``Neutrino oscillations'' occur in which one neutrino ``flavor'' (tau, muon and electron) changes into a different flavor, each carrying a specific lepton number. The primary interest in studying Neutrinos is in using standard nuclear data to determine the expected gamma ray yields from selected transitions in fission products of ^{235}U and ^{239}Pu decay chains and comparing these with our experimental data. In collaboration effort with Oak Ridge National Laboratory (ORNL), the High Flux Isotope Reactor (HFIR) and Neutron Activation Analysis facility (NAA), the gamma ray spectra of the irradiated fissile materials were obtained. Our procedure for the summer consisted of using RadiCalc (standard nuclear data program) to calculate the expected gamma ray spectrum emitted from reactor fuel samples obtained from ORNL. The predictions were then compared with the observed gamma ray spectra of RadiCalc. Preliminary work on ^{233}U , ^{235}U and ^{239}Pu was consistent with RadiCalc, within 2 standard deviations, narrowing down the possible reasons for RAA.

Methodology

In collaboration effort with Oak Ridge National Laboratory (ORNL), the High Flux Isotope Reactor (HFIR), Neutron Activation Analysis facility (NAA) and the Germanium Detector (GMX) at Temple University, the gamma ray spectra of the irradiated fissile materials were obtained. Radioactive data was extracted containing the sample count, irradiation time and thermal neutron flux at PT-2 and gamma energy spectrum for each nuclide (^{233}U , ^{235}U , and ^{239}Pu). The number of nuclide fissions after irradiation was then calculated using $N_{\text{fi}} = Y_{\text{fission}} * N_{\text{fission}}$ formula. The N_{fi} formula is further broken down as $(Nt) * (\sigma) * (\phi) * (tr)$. Where $(Nt) = ((m * Na) / M)$ and (m) being nuclide sample mass. (Na) being Avogadro's number and (M) representing molar weight of nuclide. The thermal neutron fission cross section (σ) is extracted from IAEA.org. Primarily the type of barn to be used is the “neutron cross section measured with reactor neutrons” as stated on the website. It vital to note that the unit of the barn is (x^{-24}) . The variables (ϕ) and (tr) represents thermal neutron flux at PT-2 and the irradiation time.

The sample amount was then inputted into RadiCalc. RadiCalc uses “Bateman's Laplace transform based solution to the differential equations for radioactive decay to solve for the composition and activity of radioactive sample over time” (RadiCalc: a program for estimating radiation intensity of radionuclide mixtures, 6). The time interval section was then adjusted, due to the lapse time in transportation into the reaction chamber, to “20,25,30,35,40,45,50” seconds in the multiple time selection tab. The output was then extracted into a excel file. The indicated isotopes and all their products (^{96}Y , ^{92}Rb , ^{142}Cs , ^{97}Y , ^{93}Rb , ^{100}Nb , ^{140}Cs and ^{95}Sr) were isolated in their own excel tabs for further analyzation. The maximum energy was then identified for each isotope, at each interval and then isolated as well into different excel tabs. Each isotope was then graphed from 20-50 seconds, using excel's scatter plot graphing function as seen in

figure 3. The x-axis representing time in seconds and y-axis representing “intensity (particles/second)”. From there the line of best fit equation (trendline) was used to get the area by solving $\int_{20}^{50} \text{"nuclide trendline equation"}$, then the areas are summed together. This sum represents the gamma rays being emitted per second (g/s). A manual review of the results was checked by manually calculating the Batemans equation for each beta decay chain, per nuclide. The expected results from RadiCalc and the manually calculated results were then compared for deviations.

Once again using excel's scatter plot graphing function, where the X-axis represents the channel number and the Y-axis represents gamma energy, the energy spectrum was extracted and plotted. From the plotted data each energy peak was calculated using the Peak Quantification by Non-fitting (Summation) Method. This method proceeds as followed “the non-fitting method is the primary analytic tool used to estimate sample activity. In real life gamma ray spectra, there are usually some background gamma ray peaks present even without sample. For an accurate sample count, a non-sample background gamma ray peak count needs to be subtracted from the sample count. Before any calculation is performed, sample and non-sample gamma ray peak counts are normalized by expressing them as count rates in counts per second. After subtracting the non-sample background, the continuum background in the remaining spectrum must be dealt with to obtain the net peak count rate. The non-sample count subtracted net sample count rate is given by $N' = G' - B'$ where N' is the non-sample count subtracted net peak count rate of the sample, G' is the non-sample count subtracted gross peak count rate of the sample, and B' is the non-sample count subtracted Compton continuum counts under G' . B' is determined by a trapezoid formula” (Kim 42). The expected data is then compared with the summation data for deviations and

uncertainty. The experimental method is then repeated for three radioisotopes ^{233}U , ^{235}U and ^{239}Pu .

Results

Analysis of irradiated ^{233}U , ^{235}U and ^{239}Pu samples demonstrate that the expected gamma ray intensities calculated from RadiCalc was consistent with our experimental data, within 2 standard deviations, narrowing down the possible reasons for RAA. We found that our data concluded much the same as Dr. Samuel Kim. Dr Kim stated in his findings that his results indicates that the antineutrino spectral bump at 5 to 7 MeV is not a result of errors in tabulated nuclear data. Our combined conclusion doesn't state a definite answer into the prospect of the Antineutrino Anomaly. We speculate that more refined methods of better understanding of systematic and random errors in the fission analysis software, activity estimation, energy spectrum analysis and calibration need be used. To our knowledge more experiments testing developing methods, using more samples masses and improved detectors are being conducted.

Future Prospect

The primary interest in studying Neutrinos has been using standard nuclear data to determine the expected gamma ray yields from selected transitions in fission products of ^{235}U , ^{233}U ^{239}Pu decay chains and comparing these with our experimental data. The main factors of the experiments have been heavily dependent on the energy spectrum data and its calibrations. By altering the sample amount, mainly using larger sample sizes we could extract more precise data. Allowing us to better analyze the 5-6 MeV region. The vitality of this research is in better understanding the cause for such anomalies that skews the standard model of particle physics principles. The conclusion of the Antineutrino Anomaly could bring about better understanding to limited data on neutrinos and their behaviors and assist researchers in their experiment such as the

ISLV effects to the gallium neutrino anomaly. Lastly, we aim to continue to build the foundation to understanding the mystery and behavior of complex forces much like W.I.M. P's and Dark Matter, that govern the state of our universe.

Figures

TABLE I. Most prominent beta decay branches with $E_\beta = 5\text{--}7$ MeV. The table presents the decay parent, end-point energy, half-life, and decay ft value. The decay type describes the parent and daughter states. The moderate ft values and lack of significant change of J^π suggest that all but possibly ^{140}Cs decay with allowed spectral shapes. The rate each branch contributes to the total between 5–7 MeV is N , accounting for the inverse beta decay cross section. The 1σ uncertainty due to the fission yield and branching fraction is σ_N .

Isotope	Q(MeV)	$t(s)$	$\text{Log}(ft)$	Decay type	N (%)	σ_N (%)
^{96}Y	7.103	5.34	5.59	$0^- \rightarrow 0^+$	13.6	0.8
^{92}Rb	8.095	4.48	5.75	$0^- \rightarrow 0^+$	7.4	2.9
^{142}Cs	7.308	1.68	5.59	$0^- \rightarrow 0^+$	5.0	0.7
^{97}Y	6.689	3.75	5.70	$1/2^- \rightarrow 1/2^+$	3.8	1.1
^{93}Rb	7.466	5.84	6.14	$5/2^- \rightarrow 5/2^+$	3.7	0.5
^{100}Nb	6.381	1.5	5.1	$1^+ \rightarrow 0^-$	3.0	0.8
^{140}Cs	6.220	63.7	7.05	$1^- \rightarrow 0^+$	2.7	0.2
^{95}Sr	6.090	23.9	6.16	$1/2^+ \rightarrow 1/2^-$	2.6	0.3

Sample is 301.3 nanograms Pu-239 as CRM
 Irradiated in PT-2 EOB 8/15/2016 10:27:43
 Irradiation was 30 seconds; Decay 20 seconds kept by stopwatch
 Flux was measured using Mn, Au monitors for thermal, epithermal, respectively
 Fast Flux was calculated from ORIGEN simulation taking into account the daily measured value for thermal flux
 Thermal $4.43\text{E}+13$ n/cm²/sec; Epithermal $3.24\text{E}+11$; Fast $1.11\text{E}+13$
 Counting 330.2mm sample height off the detector for 30 seconds
 Detector was Ortec 44% relative efficiency p-type coaxial with aluminum end cap
 Energy Calibration $E = 0.1807 \text{ keV} + 0.2118 \cdot \text{Ch}$
 FWHM Calibration $= 0.8052 \text{ keV} + 0.04596 \cdot E^{1/2}$

1	0
2	0
3	0
4	0
5	0
6	0
7	8
8	52
9	159
10	315
11	290
12	325
13	312
14	283
15	265
16	291
17	261

Figure A: is the gamma ray spectra of the irradiated fissile materials text file

Isotope	Energy ke	20sec	25sec	30sec	35sec	40sec	45sec	50sec	Area
Rb-92	814.98	1387016	643634.7	297921	137785.5	63707.11	29453.27	13616.52	8.86E+06
Rb-93	432.61	822128.3	454174.4	250895.7	138599.7	76565.16	42296.07	23365.17	6.80E+06
Y-96	1750.6	3027854	2110321	1470828	1025122	714478.4	497969.4	347069.3	2.91E+07
Nb-100	535.666	4529126	2781252	1707193	1047841	643137	394739.6	242280.1	4.08E+07
Cs-140	602.25	2159353	2212638	2225393	2208245	2169354	2114966	2049852	7.16E+07
Sr-95	685.6	1937947	1676351	1450067	1254328	1085011	938549.3	811858	3.37E+07
Y-97	3287.6	270146.2	107207.6	42545.35	16884.14	6700.473	2659.085	1055.258	1E+06
Cs-142	359.598	11373.89	1468.197	188.4331	24.11976	3.083546	0.393982		2.62E+04
Total									1.92E+08

Figure B: is the extracted nuclides tabulated sum of gamma rays being emitted per second (g/s)

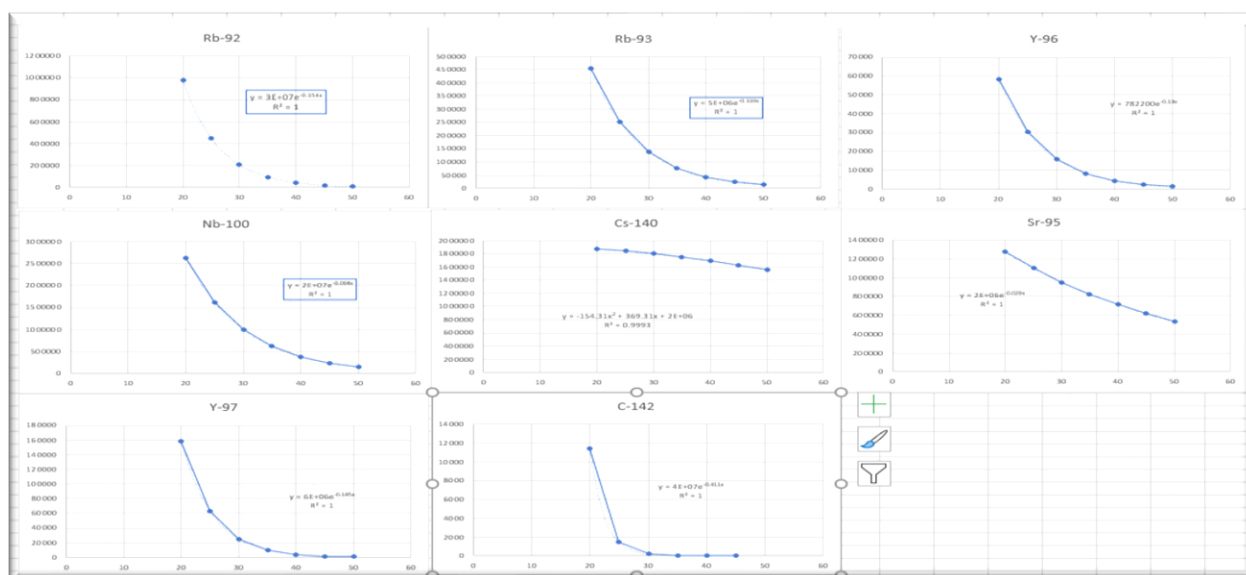


Figure C: is the nuclides graphed plots and trendline equations.

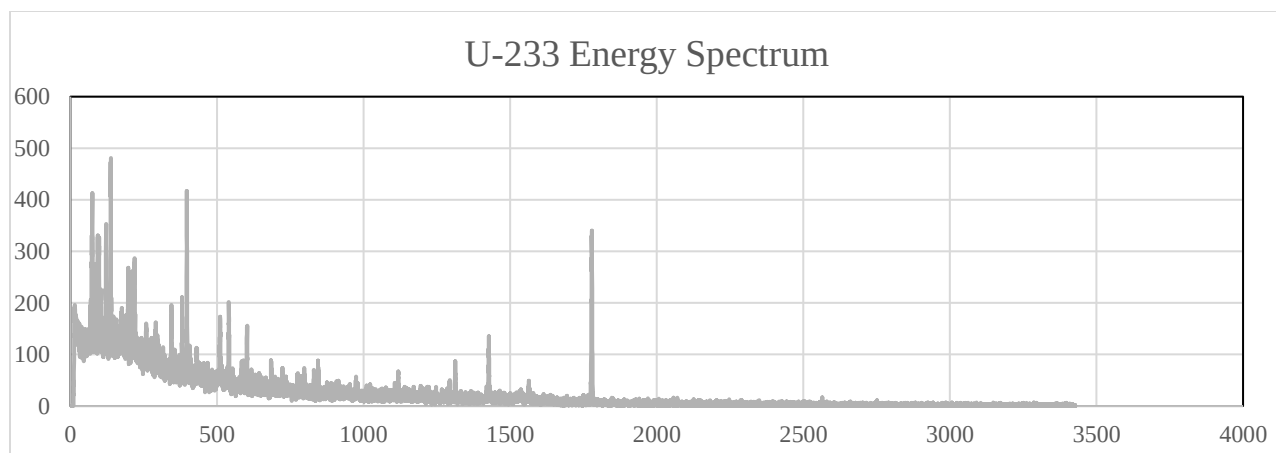


Figure D: is U-233 energy spectrum after being plotted from the irradiated fissile materials text file.

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